# Prosenjit Das

Khulna University, Khulna-9208, Bangladesh

Website Email Google Scholar GitHub LinkedIn

## Research Interests

Machine Learning & Explainable AI • Vision Transformers (ViT) • Large Language Models (LLMs) • Medical Data Analytics • Metaheuristic Optimization • Healthcare Classification

## Education

**B.Sc. in Electronics and Communication Engineering** 2023 – Present
Khulna University, Bangladesh CGPA: 3.86/4.00

## Research Experience

**Undergraduate Researcher** 2024 – Present

- Developed machine learning models for healthcare and disease classification using XGBoost, CatBoost, Gradient Boosting, and Rotation Forest.
- Applied SHAP, DiCE, and LIME for explainable healthcare prediction.
- Used metaheuristic optimization for feature selection and hyperparameter optimization.
- Worked on medical image classification using CNN and Vision Transformer architectures.
- Worked on LLM fine-tuning and evaluation.

## Journal Articles

1. **Gallstone Disease Classification Using SLOA-Optimized CatBoost Classifier with Explainable AI.**

   **Prosenjit Das**, Md. Ayaj Uddin Khan, Proshenjit Sarker, Abdullah-Al Nahid.

   *PLOS ONE.*

   Paper

2. **Dengue Fever Classification Integrating Bird Swarm Algorithm with Gradient Boosting Classifier Along with Feature Selection and SHAP–DiCE Based Interpretability.**

   **Prosenjit Das**, Proshenjit Sarker, Jun-Jiat Tiang, Abdullah-Al Nahid.

   *Applied Sciences*, Vol. 15, Article 11413, MDPI, 2025.

   Paper

3. **Aquaculture Water Quality Classification Using XGBoost Classifier Model Optimized by the Honey Badger Algorithm with SHAP and DiCE-Based Explanations.**

   S. M. Naim, **Prosenjit Das**, Jun-Jiat Tiang, Abdullah-Al Nahid.

   *Water*, Vol. 17, Article 2993, MDPI, 2025.

   Paper

4. **Breast Cancer Prediction Using Rotation Forest Algorithm Along with Finding the Influential Causes.**

   **Prosenjit Das**, Proshenjit Sarker, Jun-Jiat Tiang, Abdullah-Al Nahid.

   *Bioengineering*, Vol. 12, Article 1020, MDPI, 2025.

   Paper

5. **A Comprehensive Survey of Prompt Engineering and Context Engineering Techniques in Large Language Models.**

   Tonmoy Debnath, Md Nurul Absar Siddiky, Muhammad Enayetur Rahman, **Prosenjit Das**, Antu Kumar Guha, Muhammad Rezaur Rahman, Md. Kabir Hossain.

   *Computer Science Review*, Vol. 62, Article 100979, 2026.

   Paper

6. **Feature Selection and Hyperparameter Optimization of XGBoost Classifier Using MGO Algorithm for Gallstone Disease Classification with SHAP-Based Explanations.**

   Manuscript in preparation.

## Conference Publications

1. **Optimized Machine Learning Framework for Lifestyle and Dietary-Based Type 2 Diabetes Risk Prediction.**

   **Prosenjit Das**, Razu Sarder, Dipita Das, Anindya Nag, Hirak Mondal, Walid El-Shafai, Ahmad Taher Azar, Chakib Ben Njima.

   *International Conference on Control, Automation and Diagnosis (ICCAD 2026).*

   Paper

2. **An Explainable Hybrid CNN–Vision Transformer Framework for Lung Cancer Classification on LungHist700.**

   T. Debnath, **Prosenjit Das**, M. Rahaman, M. B. Hossain, A. K. Bairagi, A. Adhikary.

   *International Conference on Quantum Photonics, Artificial Intelligence, and Networking (QPAIN 2026).*

   Paper

3. **Aquaculture Water Quality Classification Using SLO-Optimized XGBoost with Explainable AI.**

   *International Conference on Signal Processing, Information, Communication and Systems (SPICSCON 2026).*

   Accepted and Presented.

## Datasets

1. **KU-Bangla: A Comprehensive Bangla Dataset for Language Modeling and Translation.**

   Nasif Ahmed, Khandaker Rifat, **Prosenjit Das**, Mainul Islam Labib, Abdullah-Al Nahid.

   *Mendeley Data*, Version 1, 2026.

   DOI: 10.17632/kbhwt5kjkm.1

2. **KU-BD-Fish: High-Resolution White-Background and Segmented Images for Species Classification.**

   Naim S. M., Md Ashikkur Rahman, **Prosenjit Das**, Abdullah-Al Nahid.

   *Mendeley Data*, Version 3, 2025.

   DOI: 10.17632/n9m5bywxwc.3

## Technical Skills

**Programming:** Python, C, C++, MATLAB
**ML/DL:** Scikit-learn, XGBoost, CatBoost, CNNs, PyTorch
**Vision/LLMs:** ViT-B/16, LLM Fine-Tuning, LoRA
**XAI/Optimization:** Grad-CAM, SHAP, DiCE, LIME, BSA, HBA, SLOA, MGO
**Tools:** LaTeX, GitHub

## Honors & Awards

- **1st Runners-Up** – ML Blitz Competition, IEEE Khulna University Branch.
- **University Merit Scholarship**, Khulna University, 2025. Awarded for achieving CGPA 3.88/4.00 and ranking among the top 3 students in the discipline.

## References

**Dr. Abdullah-Al Nahid**
Professor
Electronics and Communication Engineering Discipline
Khulna University, Bangladesh
Email: nahid.ece.ku@gmail.com

**Dr. Uzzal Biswas**
Professor
Electronics and Communication Engineering Discipline
Khulna University, Bangladesh
Email: uzzal.biswas@ece.ku.ac.bd